

ECEC state demand inputs: method

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1 Purpose

1. The simulator reads one file for each geography.
2. The file has six rows. Each row is one care type.
3. Each row holds one number. The number is the annual hours of child care.
4. This memo gives the source of that number.

The build covers five geographies: Kentucky (KY), New Jersey (NJ), New York (NY), New York City (NYC) and Virginia (VA).

Each number in this memo carries a tag such as [S1]. Section 5.3 lists the sources and gives the address of each one.

Two rules apply to the whole build.

- The build works at the state level. It reads county data one time, in step 1, and it adds the counties together. No later step uses a county.
- The build uses a state measurement first. It uses a national figure only where the state publishes none.

Why the build exists

- The simulator must know the mix of child care in each state.
- NSECE measures that mix [S5]. NSECE does not record the state of a family.
- NSECE reports the Census region only.
- Therefore I build the mix from the administrative records of each state.
- I use NSECE only for the care that no administrative record shows.

The position of NYC

- NYC is a geography of its own in this build. Every step below gives its figures.
- NYC is also inside NY. The NY figures include it.
- The two builds do not agree. Each build uses its own control total from step 2, and step 5 calibrates each one separately.
- Use one build or the other build. Do not add them together.

2 Results

The build measures two quantities. All other rows divide these two.

	Center-based	Not center-based	Total	Share center
KY	81,326	80,604	161,930	50%
NJ	173,225	136,721	309,946	56%
NY	270,982	378,847	649,829	42%
NYC	132,065	169,911	301,976	44%
VA	135,068	127,932	263,000	51%

Care that is not center-based has two parts. The build counts the first part. The build calculates the second part as a remainder.

	Licensed home care	Informal care
KY	4,756	75,848
NJ	1,987	134,734
NY	96,366	282,481
NYC	38,642	131,269
VA	4,743	123,189

- NY and NYC hold much more licensed home care than the other three geographies.
- This difference is correct. NY registers a home provider that keeps three children or more [S12].
- The other states do not register a provider that small. The care exists there, but no register records it.

All figures below are counts of children. They are not record counts. They are not weighted survey figures.

	KY	NJ	NY	NYC	VA
Center-Based	37,550	116,402	168,235	86,337	87,192
Unpaid Center-Based	43,776	56,823	102,747	45,728	47,876
Paid Home-Based	4,756	1,987	96,366	38,642	4,743
Other Unpaid	45,830	81,411	170,686	79,318	74,436
Other Paid	27,363	48,607	101,909	47,357	44,442
Unpaid Home-Based	2,655	4,715	9,886	4,594	4,311
Total	161,930	309,946	649,829	301,976	263,000

The output file holds annual hours. This table shows them in millions. The hours for each child come from NSECE [S5].

	KY	NJ	NY	NYC	VA
Center-Based	63.3	196.2	283.6	145.5	147.0
Unpaid Center-Based	62.9	81.6	147.6	65.7	68.8
Paid Home-Based	9.1	3.8	185.0	74.2	9.1
Other Unpaid	88.4	157.1	329.3	153.0	143.6
Other Paid	42.5	75.5	158.4	73.6	69.1
Unpaid Home-Based	5.6	10.0	21.0	9.7	9.1

3 The build

Step 1. Count the children

1. Read the Census population estimates for 2019 [S1].
2. Keep children of age 0 to 4.
3. Add the counties together for each geography.

	Counties	Children 0–4, 2019
KY	120	272,610
NJ	21	514,690
NY	62	1,127,001
NYC	5	523,718
VA	133	505,477

NYC uses five counties [S1]: Bronx 100,199; Kings 182,926; New York 76,144; Queens 137,095; Richmond 27,354. NYC holds 47 percent of the children in NY.

The build uses ages 0 to 4 for three reasons.

- The simulator uses the same ages. Its code keeps children below age 5 in seven places [S23].
- A universe of ages 0 to 5 increases each total by 15 to 23 percent [S1].
- Census publishes no county estimate for age 5 alone [S1]. The build can count a state at age 5, but it can only estimate NYC.

Step 2. Find the children in non-parental care

1. Read the NSCH household survey for 2019 [S2].
2. Keep the children of age 0 to 4. Weight each child by the child weight.
3. Calculate the share of children in non-parental care.
4. Multiply the children from step 1 by that share.

	Children 0–4	Rate	In non-parental care	Records
KY	272,610	59.4%	161,930	147
NJ	514,690	60.2%	309,946	101
NY	1,127,001	57.7%	649,829	112
NYC	523,718	57.7%	301,976	112
VA	505,477	52.0%	263,000	121

All rates and record counts in this table come from NSCH 2019 [S2]. The product is the control total. Every later count is a share of it.

- The control total is always a 2019 figure, because both inputs are 2019 data [S1, S2].
- This is why the build can use a register from 2026. The register sets the mix of care. The register never sets the total.
- NYC uses the rate of NY [S2]. NSCH publishes no rate below the state. The registers supply all NYC differences.
- This rate is the weakest number in the build. Each geography has about 120 families [S2]. The standard error is about seven percentage points. Section 5.1 explains why the build does not pool more survey years.

Step 3. Convert the registers into children

Each state licenses child care providers and publishes a register. The register gives capacity. The build needs children. Three sub-steps convert one into the other.

3.1 Remove the school-age capacity

Every register includes school-age capacity. Each register hides it differently.

- KY does not mark a provider as a center or a home [S6, S7]. The build applies the state rule: a licensed home keeps 12 children or fewer. This gives 1,609 centers and 335 homes. The build drops a provider that serves school-age children only. The build scales a mixed age band by a ratio between 0.487 and 0.827. It measures each ratio inside the register [S6].
- NJ publishes an age range for each center [S9]. The build drops the 1,035 centers that serve ages 6 to 13 only. It scales a mixed range by 0.820.
- NY publishes capacity by age [S12]. Infant plus toddler plus preschool gives the under-five capacity exactly. A home reports one total, but the register publishes its school-age capacity. The build subtracts it.
- NYC uses two registers. Its centers come from the city register [S14], which gives one capacity figure for each permit and covers children below age 6 by construction. Its homes come from the state register [S12] and take the same subtraction as NY.
- VA publishes the age range as text [S19]. The build reads the text and converts it to months. It drops a provider that starts at age 5 or later. It scales a center that serves past age 6 by 0.60.

3.2 Move the register back to 2019

Three registers are 2026 exports.

- NY scales each provider type against the state capacity series for 2015 to 2025 [S13]. Group family day care scales to 0.86. Centers scale to 0.93. One factor for all types would make the mix incorrect.
- NYC takes its own factors from the same series [S13], which publishes the city separately. Group family day care scales to 0.84 in NYC. City centers scale to 1.00.
- VA scales by one statewide factor of 0.90. The build calculates it from the state capacity report for June 2020 [S20].
- KY and NJ need no correction. The KY register is a snapshot from September 2018 [S6]. The NJ register is a snapshot from May 2019 [S9].

3.3 Convert capacity into children

Capacity is the legal maximum. The true enrollment is lower.

- KY measured its own enrollment [S8], so the build applies no fill rate there. The 2019 workforce study prints capacity and enrollment together: 42 children for each center and 9 for each home. This is the preferred method. It is why 1,609 centers give only 67,578 children.
- NJ measured its own fill rate [S10]. Its centers are 67 percent full.
- NY, NYC and VA publish no fill rate. They use the national figures: 90 percent for a center [S3] and 60 percent for a home [S4].
- NY and NYC homes use 80 percent instead [S4]. The state register already removed the school-age capacity in sub-step 3.1. The 60 percent figure would remove it a second time.

		Providers	Under-5 capacity	Children
KY	Centers	1,609	107,369	67,578
KY	Homes	335	2,115	3,015
NJ	Centers	3,128	251,159	169,106
NY	Centers, state	2,073	153,053	138,191
NY	Centers, city	2,310	141,300	127,580
NY	Homes	11,918	117,401	93,381
NYC	Centers	2,310	141,300	127,580
NYC	Homes	4,580	46,057	36,634
VA	Centers	3,186	207,835	187,654
VA	Homes	1,935	12,633	7,604

Provider counts and capacity come from the registers [S6, S7, S9, S12, S14, S19]. Children come from the conversion in sub-step 3.3.

NY needs two registers.

- NY has two regulators.
- The state licenses no center inside NYC [S12].

- NYC permits its own centers under Article 47 of the Health Code [S14].
- The state register therefore shows zero centers in NYC. This is correct. It is not a failure of the build.
- NYC uses the city register for centers and the state register for homes. The state licenses home providers in NYC.

NJ has no register of home providers. Its 1,987 children come from a survey instead [S10]: 1,396 children below age 5 across 913 responding providers, raised to the 1,254 registered providers.

Step 4. Add the programs that count their children

Head Start, Early Head Start and public pre-K publish their enrollment. The build uses the published count.

	Pre-K	Head Start	Early Head Start	Total
KY	28,465	12,307	3,004	43,776
NJ	41,305	11,940	3,578	56,823
NY	54,451	36,863	11,433	102,747
NYC	26,717	13,073	5,938	45,728
VA	33,790	11,579	2,507	47,876

- Pre-K comes from the school-level records of NCES for 2018–19 [S15].
- Head Start comes from NIEER for 2018–19 [S16]. Early Head Start comes from the NIEER report on Head Start [S17]. The KY figures agree exactly with the federal totals for 2019 [S22].
- NYC takes its Head Start figure as a share of NY [S18]: 7,194 funded places in NYC against 20,285 in the state.
- The build subtracts these children from the center count. It does not add them.
- A Head Start center usually holds a state licence. Step 3 therefore counted its children already. An addition would count them two times.
- A published count does not move in step 5. The build treats it as a census.
- This row is an upper limit in each geography. Head Start and state pre-K overlap. No source publishes the size of the overlap.

Step 5. Calibrate the total to the survey

A register measures the relative distribution across care types correctly. A register measures the total incorrectly. It shows too many children in formal care.

1. Take the share of non-parental care that is formal. NSECE gives this share for each Census region [S5]. It is 56.5 percent in the Northeast and 53.2 percent in the South.
2. Multiply that share by the control total from step 2. This gives the target.
3. Hold the program counts from step 4 constant.

4. Scale all other counts by one factor until the formal total reaches the target.

All scaled counts move together. The relative distribution does not change. Only the total changes.

	Scale factor	Formal	Informal	Informal share
KY	1.58	86,082	75,848	47%
NJ	1.04	175,212	134,734	44%
NY	1.03	367,348	282,481	44%
NYC	1.05	170,707	131,269	44%
VA	0.62	139,811	123,189	47%

This table is the primary check on the build. Read it as follows.

- NJ, NY and NYC change very little. Their register counts came within two percentage points of the survey before the build changed anything. Three separate registers [S9, S12, S14] agree with one independent national survey [S5]. This is the strongest evidence that step 3 is correct.
- The build reduces VA by one third. Its register is the most complete of the four [S19]. It lists unlicensed and voluntarily registered family homes. The other states do not record those providers. VA also uses the borrowed national fill rate [S3, S4].
- The build increases KY by one half, and this is the informative result. KY is the only state that measured how full its centers are. The measurement is 57 percent [S8]. NY and VA use 90 percent [S3]. The national figure is too high by more than one half. KY moves up and VA moves down, which is the result that this difference predicts. The calibration therefore corrects a measured error.

Step 6. Complete the rows

6.1 Use no price split

- An earlier version divided center care into low-priced and high-priced.
- A state publishes one price [S21]. Each state therefore fell on one side of the national boundary.
- A care type with no children then reads as unmeasured, not as absent.
- Center care is now one row. It holds 1,686 hours for each child each year. This is the national mean across both price bands [S5].

6.2 Calculate the care that no register records

- Three of the six types appear in no register: unpaid relatives, nannies and informal arrangements.
- Unpaid relative care is not a business. It files no return.
- The build subtracts all counted care from the control total. The remainder is this care.
- The build divides the remainder three ways on the national pattern [S5].
- The remainder is 44 to 47 percent of non-parental care. The national figure is 47 percent [S5].

6.3 Convert children into hours

- Each care type has a measured hours figure from NSECE [S5].
- The figure is the mean weekly hours of the primary arrangement, multiplied by 50 weeks.
- The figures run from about 1,440 hours for program center care to about 2,120 hours for unpaid home care [S5].
- The hours column is therefore not proportional to the children column.

4 The source of each number

	Source	Reliability
Children 0–4	Census, 2019 [S1]	Counted
Share in non-parental care	Household survey, 2019 [S2]	Weak. About 120 families for each geography. The build tested a pool of survey years and rejected it
Provider capacity	State registers [S6, S7, S9, S12, S14, S19]	Counted, but it is a legal maximum. VA providers report that they can serve 27 percent fewer children than their licence permits [S21]
Fill rate, KY and NJ	The state’s own survey [S8, S10]	Measured in the state
Fill rate, NY, NYC and VA	National survey [S3, S4]	Borrowed. Two states report that it is too high [S8, S21]
Program enrollment	Federal and state reports [S15, S16, S17, S18]	Counted
Informal split and hours	National survey [S5]	Borrowed

5.1 Why the rate uses one survey year

A pool of survey years reduces the sampling error by 40 to 47 percent [S2]. The build does not pool. Every input must match the 2019 base year of the simulator.

1. A pool of 2018, 2019 and 2020 centres exactly on 2019. This pool passes the vintage test.
2. This pool fails on content. The survey collected the 2020 data during the pandemic [S2]. The use of child care fell.
3. A pool of 2018 and 2019 centres on the middle of 2018. Every other input is 2019.

	2018	2019	2020
KY	57.3%	59.4%	45.8%
NJ	58.3%	60.2%	42.6%
NY	64.7%	57.7%	62.1%
VA	61.7%	52.0%	46.7%

All figures in this table come from NSCH [S2]. The fall in 2020 is a real change. It is not sampling noise. A pool would decrease each total by several percentage points. The rate therefore keeps its full sampling error. The repository holds all three survey years, so another person can repeat this test.

5.2 Why the borrowed fill rates are too high

Two states published their own figures. Both point in the same direction.

- KY measured its centers at 57 percent full in 2019 [S8]. NY, NYC and VA use 90 percent [S3].
- VA measured a related quantity in 2022 [S21]. It asked 1,968 providers what they can serve, and compared the answer to their licence. Centers reported 73 percent of their licensed capacity. Homes reported 91 percent. The sample holds 121,260 licensed places against 89,374 children.

The build uses neither figure as a rate.

- The KY figure applies to KY.
- The VA figure is from 2022. It measures capacity, not enrollment. It would duplicate the correction that the fill rate already makes.
- Both figures show the direction and the size of the error. Licensed capacity is too high by one quarter to one third. The national fill rate does not remove enough of it.

5.3 Source list

- S1** Census Bureau, Population Estimates, county characteristics, vintage 2019. Children 0–4 for each county.
<https://www2.census.gov/programs-surveys/popest/datasets/2010-2019/counties/asrh/cc-est2019-alldata.csv>
- S2** Census Bureau, National Survey of Children’s Health, topical files for 2018, 2019 and 2020. Variable `k6q20`, weighted by `fwc`.
<https://www.census.gov/programs-surveys/nsch/data/datasets.html>
- S3** Administration for Children and Families, *Center-Based Early Care and Education Providers in 2012 and 2019*. Center fill rate of 90.29 percent, from the five age bands below age 5.
<https://acf.gov/opre/report/center-based-early-care-and-education-providers-2012-and-2019-counts-and>

- S4** Administration for Children and Families, *Home-Based Early Care and Education Providers in 2012 and 2019*, May 2021. Home fill rate of 79.54 percent and under-five share of 75.67 percent.
<https://acf.gov/opre/report/home-based-early-care-and-education-providers-2012-and-2019-counts-and-characteristics>
- S5** National Survey of Early Care and Education 2019, public-use files, ICPSR 37941. Formal share by region, informal split, hours for each care type, and the price bands behind the 1,686-hour figure.
<https://www.icpsr.umich.edu/web/ICPSR/studies/37941>
- S6** Child Care Aware of America, KY licensed provider register, September 2018.
https://services7.arcgis.com/s3vGpGobX9nzlLH3/arcgis/rest/services/Providers_Licensed/FeatureServer
- S7** Child Care Aware of America, KY certified home provider register, September 2018. Same service, layer `Providers_Certified`.
- S8** Child Care Aware of Kentucky, *2019 Kentucky Workforce Study*, Table 20. Capacity and enrollment by age band.
<https://www.childcareawareky.org/research/>
- S9** NJ Department of Children and Families, licensed child care centers, May 2019.
<https://data.nj.gov/resource/cru5-4rmm.csv>
- S10** NJ, *2021 Child Care Market Rate Survey*. Center fill rate of 67.33 percent, and the family child care provider universe.
<https://www.childcarenj.gov/Resources/Reports>
- S12** NY Office of Children and Family Services, licensed child care providers.
<https://data.ny.gov/resource/cb42-qumz.csv>
- S13** NY Office of Children and Family Services, provider counts and capacity by modality, 2015 to 2025. Published separately for NYC and the rest of the state.
- S14** NYC Department of Health and Mental Hygiene, child care centers permitted under Article 47.
<https://data.cityofnewyork.us/resource/gy3q-4tzip.csv>
- S15** National Center for Education Statistics, Common Core of Data, school-level pre-kindergarten membership, 2018–19.
https://nces.ed.gov/ccd/Data/zip/ccd_sch_052_1819_l_1a_091019.zip
- S16** National Institute for Early Education Research, *The State of Preschool 2019*.
<https://nieer.org/state-preschool-yearbook-2019>
- S17** National Institute for Early Education Research, *The State(s) of Head Start and Early Head Start*, 2023.
https://nieer.org/sites/default/files/2023-10/hs_fullreport.pdf
- S18** Administration for Children and Families, Head Start service locations and funded places.
https://s3foa.s3.us-east-1.amazonaws.com/HS_Service_Locations.csv

- S19** VA Department of Social Services, licensed and approved child care providers.
<https://legacy.dss.virginia.gov/facility/search/cc2.cgi>
- S20** VA, *Child Care Data Matrix: estimated child care capacity and demand*, June 2020.
<https://data.virginia.gov/dataset/child-care-data-matrix-virginia-s-estimated-child-care-capacity-demand-june-2020>
- S21** Miller-Bains, Katherine, Stephen Yu and Daphna Bassok, *Is Authorized Capacity a Good Measure of Child Care Providers' Current Capacity? New evidence from Virginia*, EdWorkingPaper 24-983, 2024.
<https://edworkingpapers.com/ai24-983>
- S22** Administration for Children and Families, *Head Start Program Facts, Fiscal Year 2019*. Used to check the KY program figures.
<https://acf.gov/ohs/data/program-facts>
- S23** ECEC-Simulator, `src/1_processing/1b_acs_processing.R`. The filter `age < 5` appears in seven places.

5 Limits

1. The build cannot see unpaid relative care. This is the largest informal category [S5]. The figure is a remainder, not a measurement. No administrative source would correct it. Only a restricted survey file would.
2. The mix carries the date of the register. The total is 2019 by construction. The division across care types comes from the last publication of the state. The mix changes over time. A register from 2026 shows too few family homes for 2019.
3. Unpaid Center-Based is an upper limit in each geography. Head Start and state pre-K overlap [S16, S17]. A program site can also hold a state licence.
4. Step 5 forces an exact match to the survey. This is a decision, not a result. The alternative is to report the disagreement between the register and the survey.
5. The simulator expects seven rows. It divides center care by price, and it rejects a file with six rows [S23]. The bridge between the two is one mechanical operation. It divides the center row on a national share. The build keeps this operation outside the method, so that the build does not appear to measure a quantity that it cannot measure.